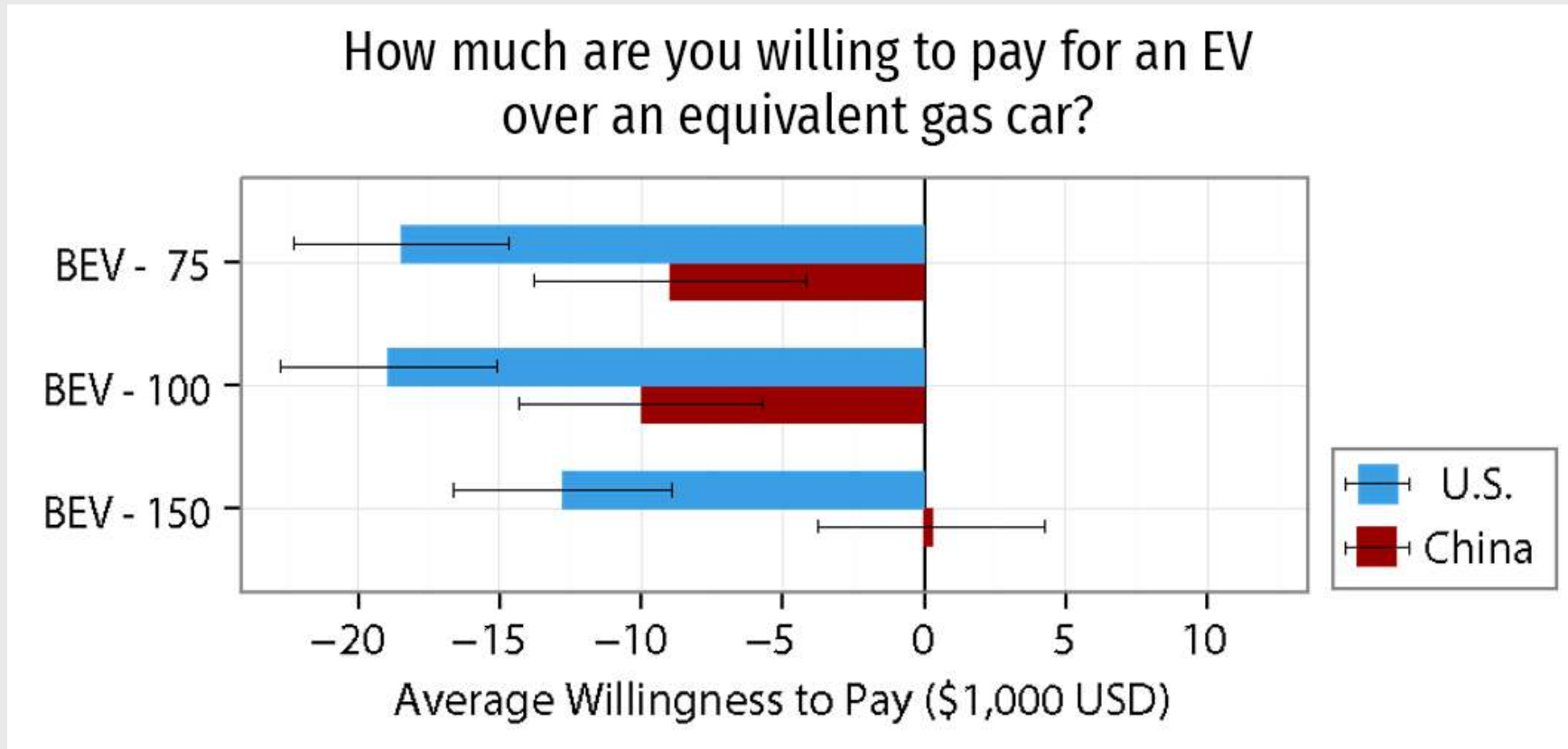


Human choices and the energy transition: Barriers to electric vehicle adoption

John Paul Helveston, George Washington University

IES Energy Symposium 2026

US buyers are more opposed to EVs than Chinese buyers

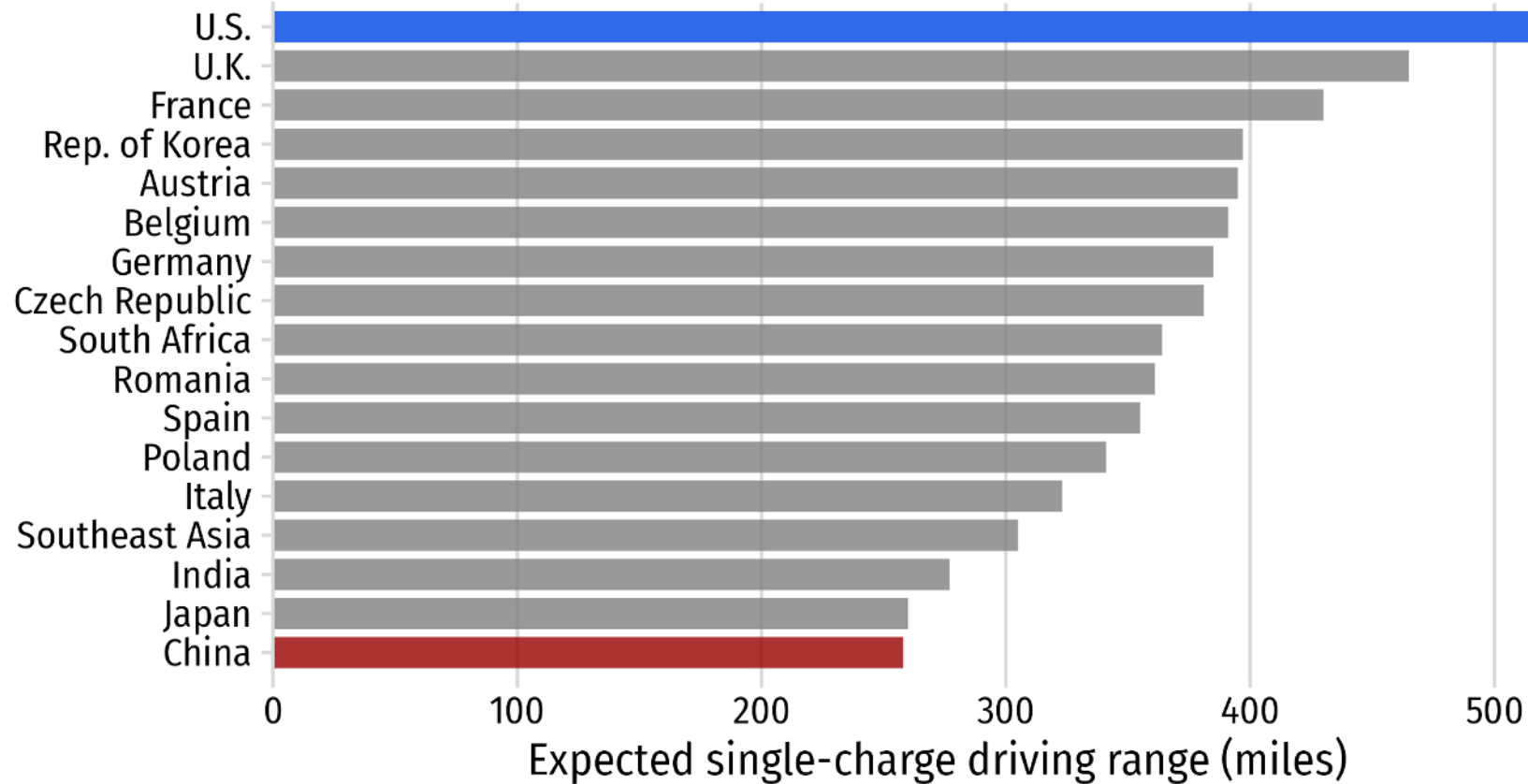


Helveston et al. (2015) "Will subsidies drive electric vehicle adoption? Measuring consumer preferences in the U.S. and China" *Transportation Research Part A: Policy and Practice*. 73, 96–112. DOI:

[10.1016/j.tra.2015.01.002](https://doi.org/10.1016/j.tra.2015.01.002)

US buyers demand far more driving range than Chinese buyers

Average response to question: "How much driving range would a fully charged all-battery electric vehicle need to have in order for you to consider acquiring one?"



Data source: Deloitte 2022 Global Automotive Consumer Study

Top 4 Selling Vehicles in **China**

Xiaomi SU7

\$30,171 - \$41,909



BYD Song

\$24,721 - \$38,555



Geely Xingyuan

\$ 9,615 - \$13,667



Tesla Model Y

\$36,822 - \$43,809



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Top 4 Selling Vehicles in **USA**

Toyota RAV4

\$29,250 - \$38,955



Honda CRV

\$30,100 - \$41,100



Ford F-150

\$44,095 - \$79,005



Chevrolet Silverado

\$42,700 - \$70,000





OPINION
GUEST ESSAY

Where Did All the Affordable Cars Go?

 Share full article



 3.2K

By Clifford Winston

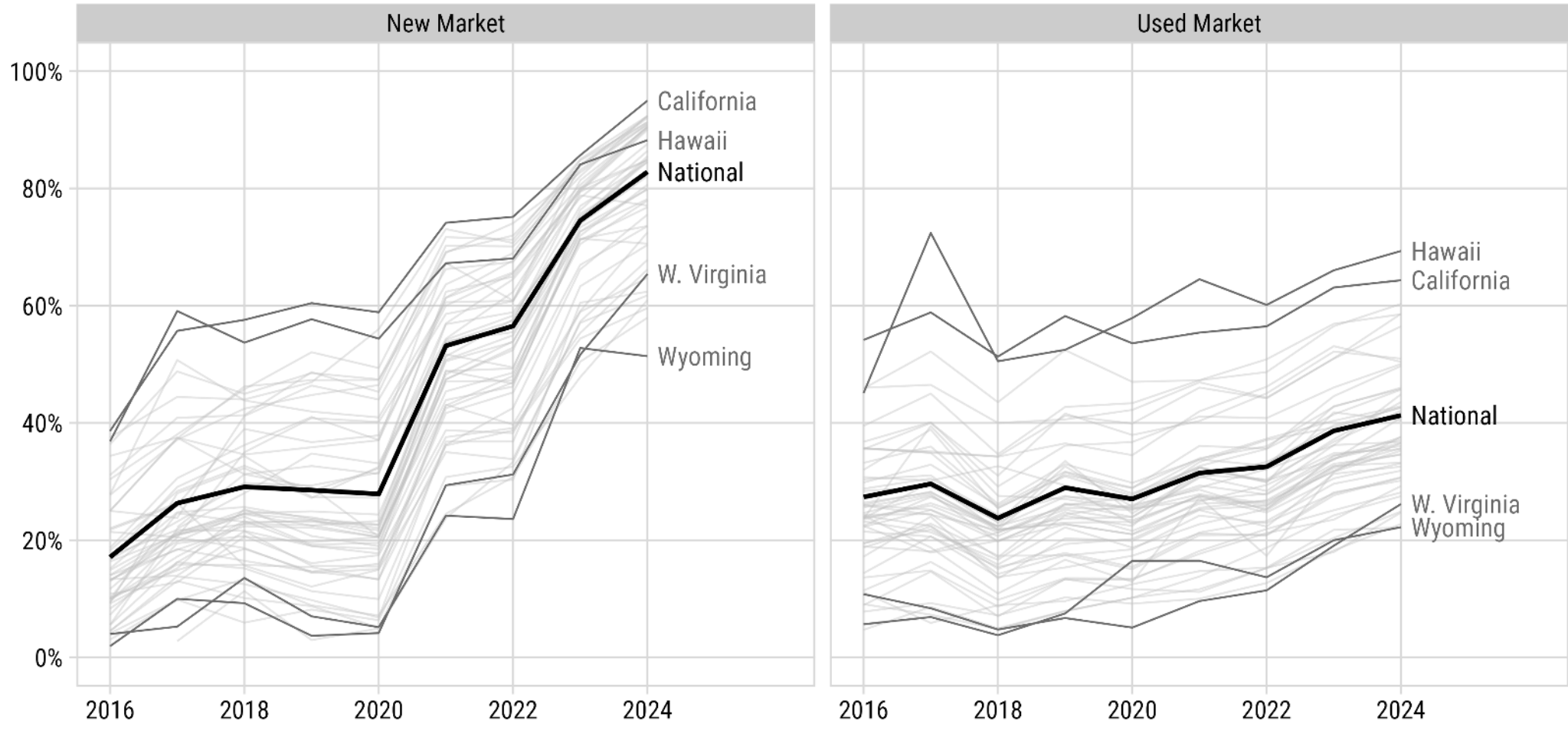
Mr. Winston is an economist who specializes in transportation and microeconomic policy.

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How many dealerships are carrying EVs?

4/5 dealers have *new* BEV; 2/5 dealers have *used* BEV

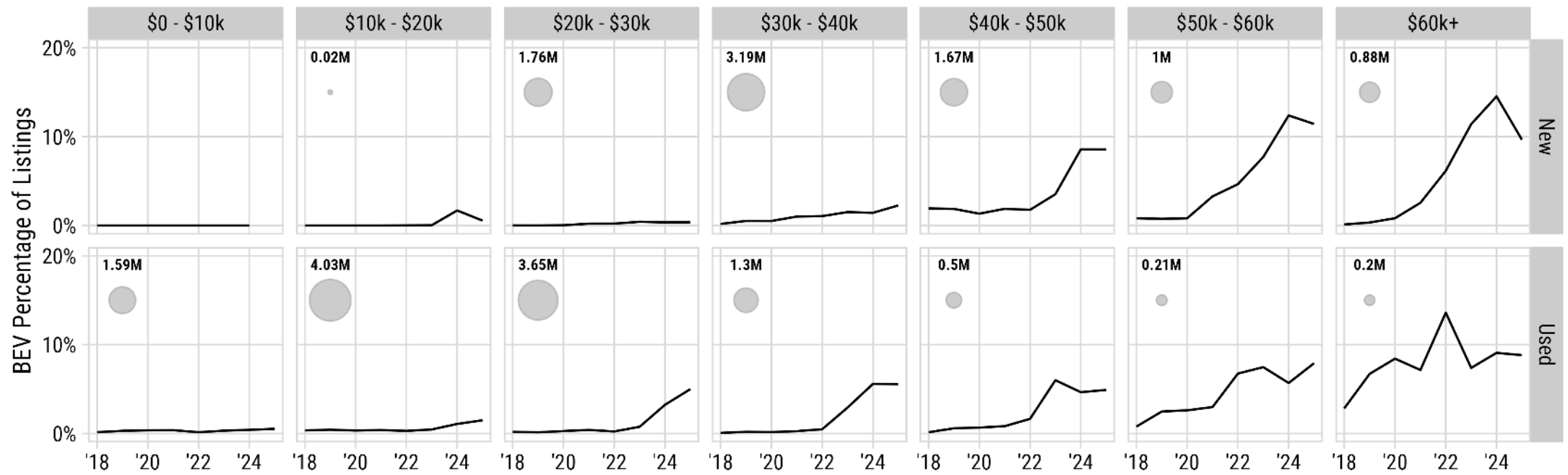
Percentage of dealerships with at least one PEV listing



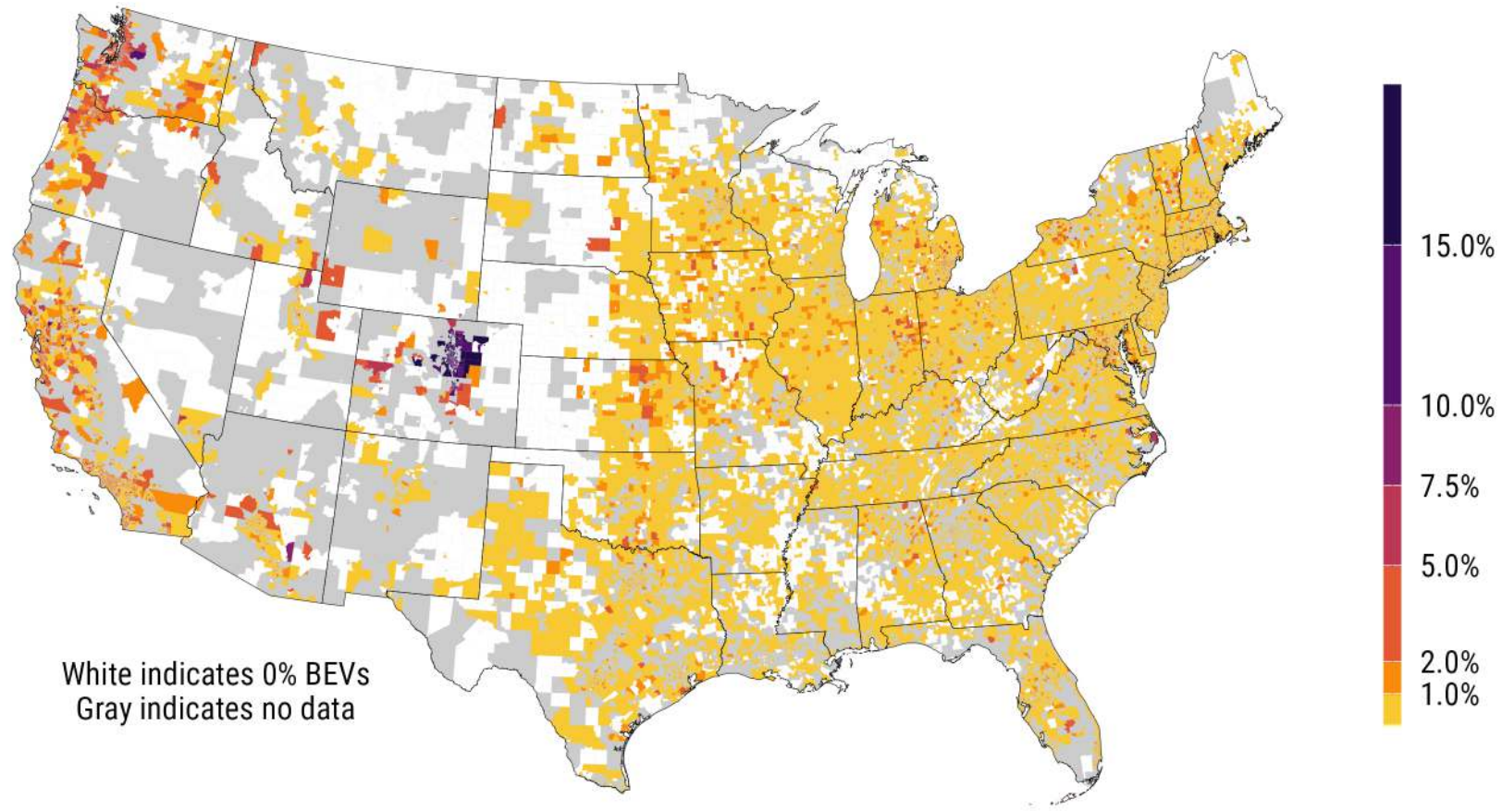
Majority of BEV listings in high-price segments

BEV percentage of vehicle listings in new and used markets by price bin

Bubbles show total 2025 listings (in millions) for each market and price bin



The EV "Deserts" of America



Data: **128 M vehicle listings** from 60k dealerships (2016 - 2025), marketcheck.com

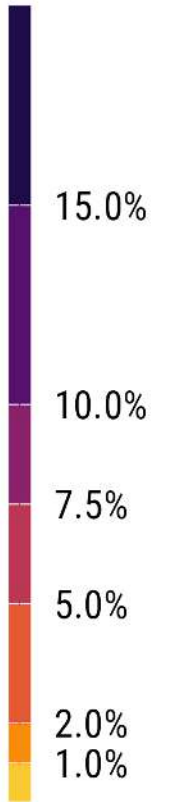
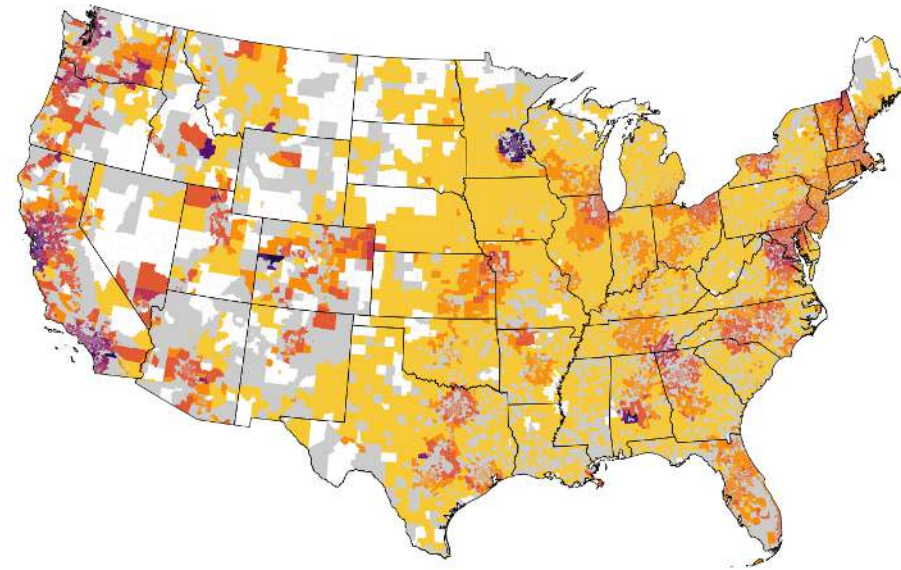
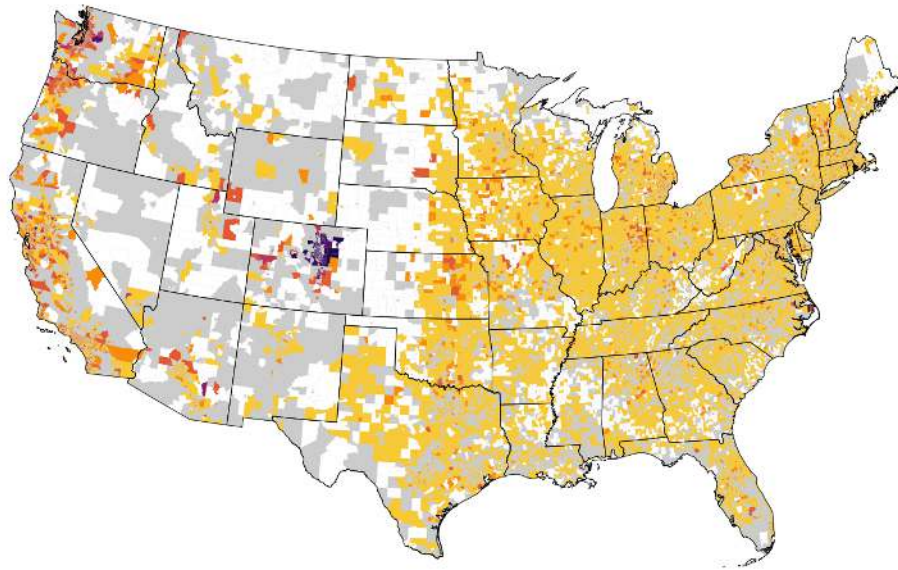
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The EV "Deserts" of America

White indicates 0% BEVs
Gray indicates no data

New

Used



Data: **128 M vehicle listings** from 60k dealerships (2016 - 2025), marketcheck.com

Thanks!

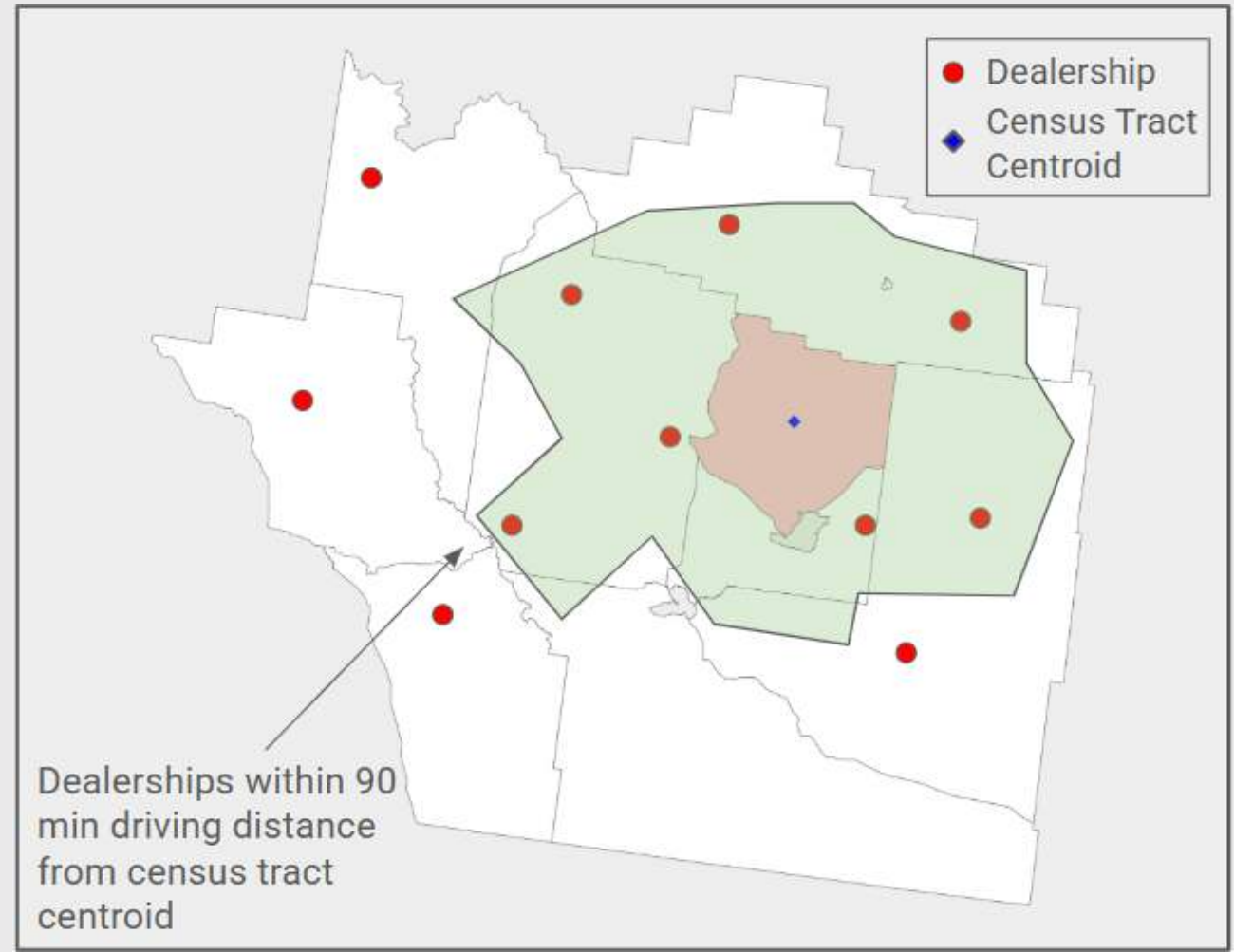
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Extra Slides

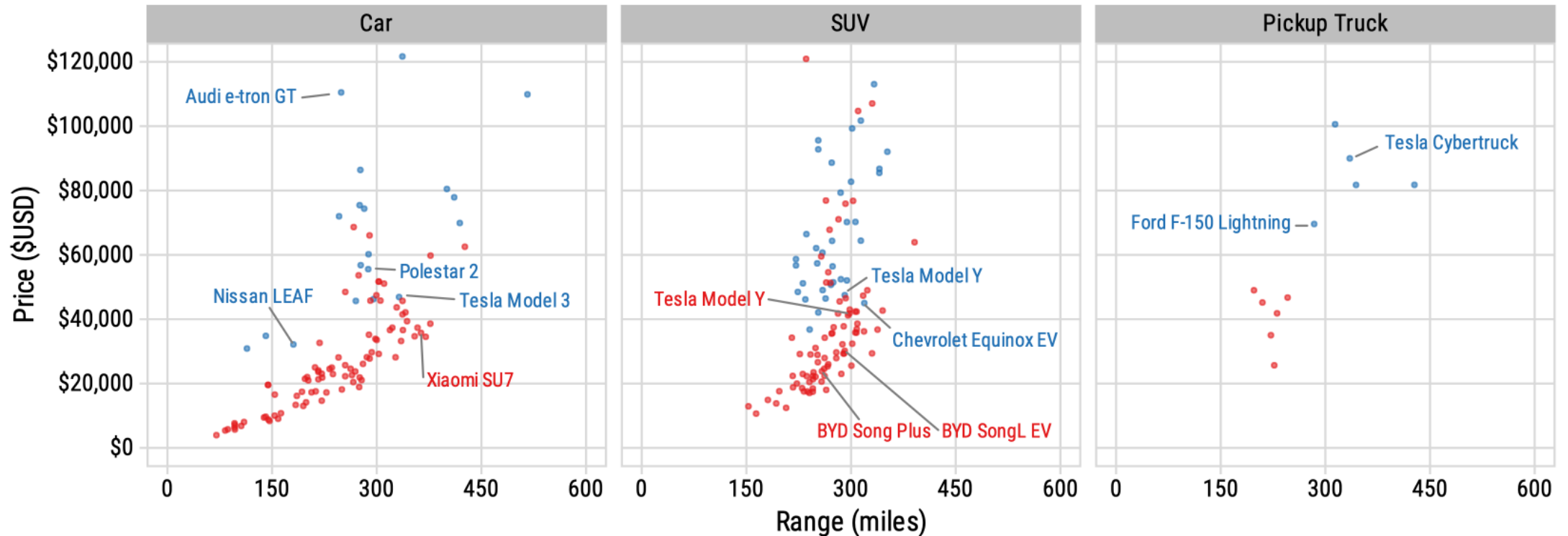
Local vehicle market: Vehicles in 90 minute driving isochrone

Combine ML algorithm of predicted household vehicle budget (using Census data) with local market characteristics



China offers more affordable BEVs across all range categories

Price vs. Range for all Model Year 2024 BEVs in **China** and the **USA**



Data scraped from autocango.com (China) and carsheet.io (USA)

Interactive version at <https://jhelvy.github.io/science-2025/>